

Earth Embeddings for Political Science

How They Work, What They Measure, What Comes Next

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Three questions the talk will try to answer.

1. How do we capture a *place* in a number?
2. What can we do with that number to recover information about *similar* places?
3. How do we solve the puzzle that two places so close can be *worlds apart*?

Political science asks spatial questions all the time.

How does a policy diffuse across jurisdictions? Where does the state invest, and where does it withdraw? Why do some neighborhoods receive public services and others do not?

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But the tools we use to *represent a place* are remarkably crude:

A row in a panel

country-year,
unit-year, one
observation at a time

**A polygon
boundary**

a municipality, a dis-
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**A distance or
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a dummy, a kilometer,
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What a place *actually looks like on the ground* — its built environment, its roads, its land use — almost never makes it into the equation.

Earth Embeddings as a Possible Answer

- Vector representations of geographic locations, produced by deep neural networks trained on satellite imagery.

For each location on Earth

$[0.31, -0.47, 0.12, \dots, 0.08]$

$$\mathbf{e} \in \mathbb{R}^{64}$$

AlphaEarth Foundations, 2025

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- A single **64-number vector** per location encodes built environment, road network, vegetation, land use, spatial organization.
- **Query by coordinate.** Merge with any georeferenced dataset. Feed into any standard model.
- A **very recent** object of research, with **essentially no uptake** in political science so far.

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PART 1

How They Work

The embedding concept,
the production pipeline,
and the model ecosystem



PART 2

What They Measure

Where embeddings fit in
the longer tradition of
satellite-based measure-
ment in social science



PART 3

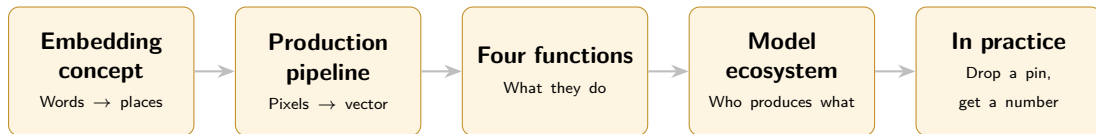
A Methodolog- ical Opening

Embeddings as a new ingre-
dient for spatial estimation

Parts 1 and 2 review an emerging technology. Part 3 sketches where the conceptual opening lives.

Part 1: How Earth Embeddings Work

The technology, from satellite pixels to a 64-dimensional vector



Question answered in Part 1:

How do we get from a satellite image to a short vector?

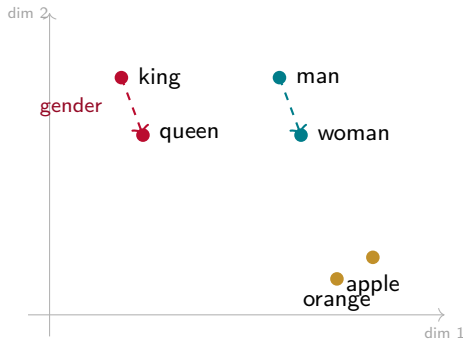
The Embedding Concept — Starting with Words

An embedding represents a complex object as a point in a vector space, arranged so that **geometric relationships** between points reflect **substantive relationships** between objects.

The canonical example: **word2vec** (Mikolov et al., 2013).

- “king” and “queen” end up close in vector space
- “king” – “man” + “woman” $\approx$ “queen”
- Words in similar contexts end up near each other

The structure a model internalized about language *emerges* as geometric structure in the embedding space.



Structure learned from the data shows up as *direction* and *distance* in vector space.

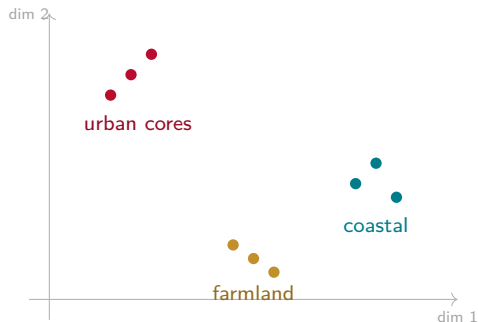
From Words to Places

Earth embeddings apply the same logic to geography.

An embedding function E maps a (location, time) pair into a d -dimensional vector:

$$E : (\text{location}, \text{time}) \rightarrow \mathbf{e} \in \mathbb{R}^d$$

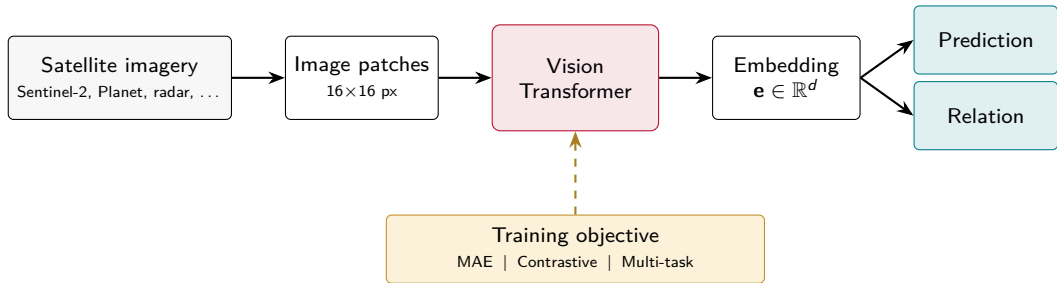
- Embedding distance summarizes similarity in visible surface structure (e.g., urban form), not map distance.
- The figure shows this geometry: urban cores cluster, while coastal and farmland patterns separate.



Geographic similarity emerges as geometric clustering in embedding space.

An earth embedding is a **learned summary of what a location looks like from above**, compressed into a **small set of numbers**.

The Production Pipeline — Three Stages



Three design choices shape what the embedding *knows*:
input data · model architecture · training objective.

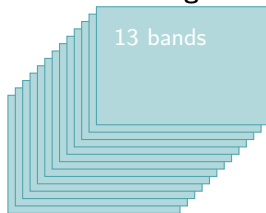
Stage 1: Input Data

Models are trained on geospatial observations indexed by location and time. The dominant input is **multi-spectral satellite imagery**.

- **Sentinel-2** · 10-meter, 5-day revisit, 13 bands, free
- **Planet** · 3–5 meter, daily, commercial
- **Sentinel-1** radar · cloud-robust
- Auxiliary layers: POIs, demographics, geotagged text

Input resolution sets the *finest scale* at which embeddings can distinguish places; spectral richness sets *what kinds of features* they can encode.

A satellite image is not a JPEG.



A satellite image is a

raster: a grid of cells where each cell records light intensity in several wavelengths. A Sentinel-2 scene has thirteen bands, including near-infrared and short-wave infrared, which are the bands that tell you about vegetation moisture, mineral content, and other things the human eye cannot see.

Stage 2: The Vision Transformer

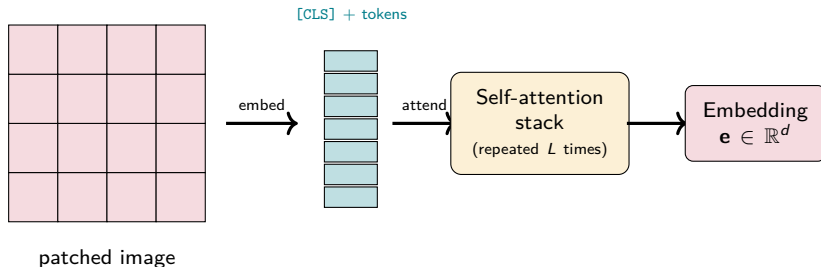
The dominant architecture. Dosovitskiy et al. (ICLR 2021): “*An Image is Worth 16×16 Words.*”

- 1 Split the image into fixed patches (e.g., 16×16).
- 2 Linear projection: each patch $\rightarrow$ a vector token.
- 3 Add a *positional encoding* so the model knows where each patch sits in the image.
- 4 Stack of transformer blocks: **self-attention** + MLP + residual connections.
- 5 Output: contextualized patch tokens plus a special [CLS] token (*classification token*) that aggregates the whole image.
- 6 [CLS] (or a pooled combination of tokens) *is* the embedding.

Self-attention lets every patch look at every other patch *directly*: a patch with ambiguous texture is interpreted using evidence from many parts of the scene, not just immediate neighbors. **Patch meaning comes from scene-level context.**

CNNs (ResNet, EfficientNet) remain viable, but transformers have become the default.

The Vision Transformer — Visually



Patches become tokens. Tokens attend to each other across the whole image. The [CLS] token ends up carrying a summary of the scene — that summary is what we call the embedding.

Stage 3: Training Objectives — What Does the Embedding Learn?

Objective	Mechanism	What the embedding learns	Example model
Masked autoencoding	Hide $\sim 75\%$ of patches; re-construct them. Loss: $\ x - \hat{x}\ ^2$	Visual texture, spatial regularities of landscapes	Clay, Prithvi
Contrastive	Pull aligned (coord, image) pairs together; push random pairs apart	Location identity from recurring observable patterns	SatCLIP, GeoCLIP
Supervised multi-task	Predict several known targets (land cover, temperature, elevation, ...) from a shared backbone	Whatever is jointly predictive of the supervised bundle	AlphaEarth

Same imagery, different objective $\Rightarrow$ **different geometry** and **different downstream uses**.

Brown et al. (2025), Google DeepMind. The model this paper returns to throughout.

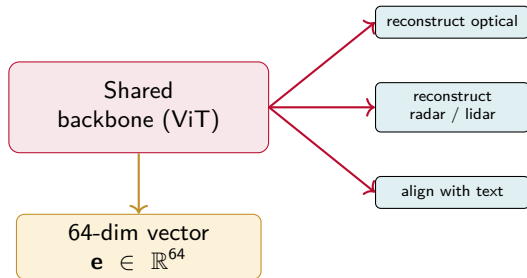
Inputs (multi-source)

- Sentinel-2 optical, Sentinel-1 radar, Landsat
- GEDI lidar, climate reanalysis
- Geo-tagged Wikipedia text

Training (simultaneously)

- Reconstruct each input source from the embedding
- Teacher–student contrastive alignment, plus alignment to geo-tagged Wikipedia text

The 64-dim vector has to be predictive of *all* of these signals at once.



The applied researcher does not train this model. The product is a pre-computed catalog, queryable on Google Earth Engine.

More detail: [AlphaEarth \(paper detail\)](#)

What Earth Embeddings Do — Four Functions (Klemmer et al., 2025)

Compression

Distill high-dimensional, multi-modal geospatial data into compact vectors. A satellite patch with millions of pixels becomes 64 numbers.

Interpolation

Return embeddings for *any* coordinate in continuous space — including locations absent from training data. Enables coordinate queries.

Fusion

Combine optical imagery, radar, text, and demographic data into a single representation indexed by location. One pipeline, not five.

Integration

Serve as the standard geographic token for other AI systems, the way word embeddings became the standard input for language models.

For political scientists today: **compression** and **fusion** are the immediately useful pair. Interpolation and integration are about to matter more.

The Model Ecosystem

Model	Input	Training	Dim.	Access
AlphaEarth	Sentinel-1/2 + auxiliary	Multi-task	64	Google Earth Engine
Clay	Multi-sensor imagery	MAE	varies	Open weights
Prithvi	Harmonized Landsat–Sentinel	MAE	768	Open weights (NASA + IBM)
SatCLIP	Coords ↔ Sentinel-2	Contrastive	256	Open weights
GeoCLIP	Coords ↔ ground-level	Contrastive	varies	Open weights

What an Embedding Looks Like in Practice

Drop a pin on Earth. Ask the model:

“what does this place look like, as a number?”

- Pin over your head now $\rightarrow \mathbf{e}_1$
- Pin on the street you grew up on $\rightarrow \mathbf{e}_2$
- Pin on your dream beach $\rightarrow \mathbf{e}_3$

Each vector is 64 floating-point numbers.
Three short rows. That is the object.

You can compare them with standard vector-distance or similarity metrics, merge them with a census table, or feed them into a regression just like any other covariate block.

AlphaEarth query result (abbreviated)

WashU	38.648, -90.305	[0.31, -0.47, ..., 0.08]
Mexico City	19.432, -99.133	[0.02, 0.55, ..., -0.04]
Rocinha	-22.983, -43.232	[-0.18, 0.09, ..., 0.27]

Three vectors in $\mathbb{R}^{64}$.

The entry cost is a coordinate query. The output is a vector. The shape of the workflow looks exactly like any other tabular analysis.

Earth embeddings compress the observable character of a place into a short vector,
produced by Vision Transformers trained with self-supervised objectives on satellite imagery.

Input

Multi-spectral satellite rasters,
radar, auxiliary layers.

Architecture

Vision Transformer — patches →
self-attention → a [CLS] vector.

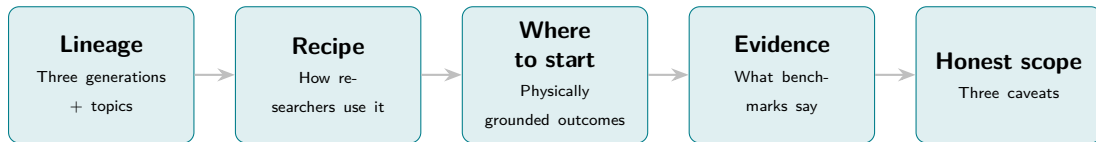
Objective

MAE, contrastive, or supervised
multi-task — each shapes a
different geometry.

*With the machinery out of the way: **what can these vectors actually measure?***

Part 2: What Earth Embeddings Measure

Prediction, measurement, and where political science should start



Question answered in Part 2:

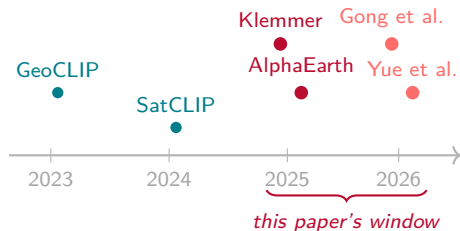
Where does this technology fit in the applied empirical toolkit, and where does the evidence say it actually works?

Before We Review the Evidence: How New Is All This?

Earth embeddings are, as of this talk, a **very young technology**.

- Klemmer et al. (2025): framework paper, formalizes the category — **2025**
- AlphaEarth Foundations (Brown et al.): flagship queryable database — **2025**
- SatCLIP, GeoCLIP: slightly earlier, narrower — **2023–2024**

No peer-reviewed political-science paper currently uses earth embeddings as defined here. The gap is *open*, not ignored.



The category, as an object of research, is less than three years old.

Three Generations of Satellite-Based Measurement

Generation	Approach	Canonical example	Limitation
Hand-crafted proxies	One interpretable index tied to one outcome	Henderson, Storeygard & Weil (2012): nighttime lights $\rightarrow$ GDP	New outcome $\Rightarrow$ new proxy
Task-specific deep learning	CNN trained per task; intermediate features reused	Jean et al. (2016): poverty in Africa	Pipeline rebuilt per outcome, per region
General-purpose embeddings	Pre-computed vectors usable across many outcomes	AlphaEarth, SatCLIP, Clay (2023–2025)	Construct validity, interpretability, spatial bias

Each generation **lowers the cost** of turning imagery into a usable feature for applied work — from a custom proxy per outcome, to a custom model per task, to a single pre-computed vector reusable across many.

What's Been Done — and What's Missing

Economic & welfare

- ✓ Country & subnational GDP from nightlights
Henderson, Storeygard & Weil (2012)
- ✓ Consumption & asset wealth in 5 African countries
Jean et al. (2016)
- ✓ Country GDP from earth embeddings
Yue, Zhao & Hu (2026)
- ✗ No serious political-science test of whether embeddings improve on these.

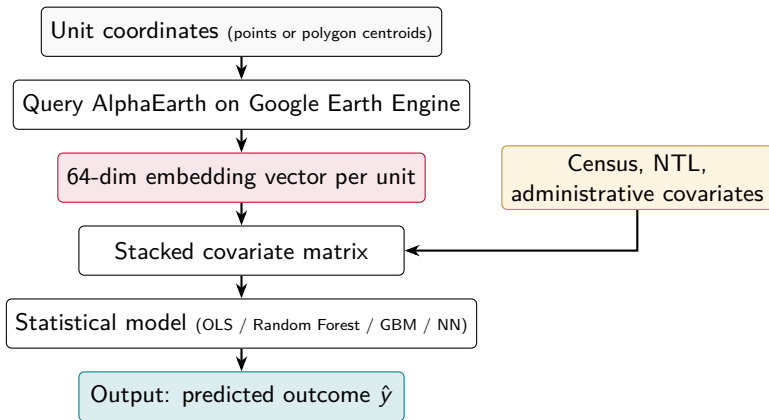
Population & built environment

- ✓ Global gridded population estimates
Tatem (2017, *WorldPop*)
- ✓ Slum detection from high-resolution imagery
Mahabir et al. (2018)
- ✓ Urban indicators in 6 U.S. metros from embeddings
Gong et al. (2026)
- ✗ Cross-country evidence concentrated in the Global North.

Political-science outcomes

- ✓ Remote conflict monitoring
Sticher, Wegner & Pfeifle (2023)
- ✓ Political violence from fire data
Naghizadeh (2024)
- ✓ Human-rights documentation
Edler et al. (2025)
- ✓ Public-service accessibility
Huang et al. (2024)
- ✗ None of these use earth embeddings yet.

An Applied Workflow in the Earth-Embedding Era



No image processing. No deep-learning infrastructure.
The entry cost has dropped to a **coordinate query**.

Given how earth embeddings are constructed, the natural place to start in political science is outcomes that **leave a physical trace on the ground.**

Public goods, broadly.

- Roads, transit, electricity grids
- Schools, hospitals, water, sanitation
- Public safety infrastructure

Other physical infrastructure.

- Housing stock, commercial density
- Port activity, transport networks
- Informal settlement growth, urban form

Almost all of these leave a visible signature from orbit. A state that paves roads in some places and not others is, quite literally, drawing its priorities into the landscape.

And Where the Tool Buys the Most Information

In the settings where this kind of measurement matters most, **conventional data is exactly what we don't have.**

Settings where the gap is real.

- Low- and middle-income countries
- Under-administered regions
- Informal neighborhoods and unrecognized settlements
- Post-conflict and post-disaster zones

What is typically missing.

- Recent or reliable censuses
- Continuous administrative records
- Affordable, high-resolution survey coverage
- Up-to-date cartography of the built environment

A queryable representation of *what a place looks like* is sometimes the **only data source available at comparable spatial resolution.**

Gong et al. (2026) make this point explicitly.

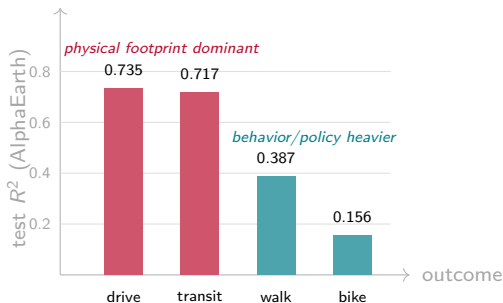
What the Evidence Says — Gong et al. (2026)

The strongest systematic benchmark so far.

- Compares AlphaEarth, Prithvi, Clay
- 14 neighborhood-level urban indicators
- 6 U.S. metropolitan areas, 2020–2023

Central findings.

- AlphaEarth's 64-dim vectors beat the higher-dim alternatives
- Skill strongest for outcomes tied to **built-environment structure** (commuting modes, urban form)
- Skill weakest for outcomes driven by **fine-scale behavior or policy** (cycling rates, fine local variation)



Best where physical features dominate
— exactly the physically grounded space.

Values shown are AlphaEarth test R^2 from Gong et al. (2026),

Supplementary Table 2.

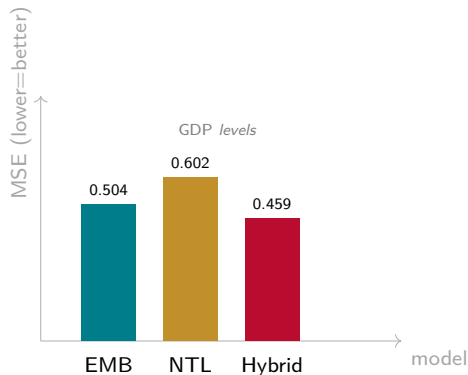
What the Evidence Says — Yue, Zhao & Hu (2026)

Country-level GDP estimation,
AlphaEarth + nighttime lights. ~ 153 countries $\times$
8 years (2017–2024); 1,225 country-year observations.

- 1 Train small network: embeddings $\rightarrow$ NTL
- 2 Truncate at a hidden layer $\Rightarrow$ “income-aware” 32-dim representation $\mathbf{z}_i$
- 3 Feed $\mathbf{z}_i$ into a GDP predictor

Headline numbers.

- **Cross-sectional GDP levels:** embeddings cut MSE by $\sim 16\%$ vs. NTL alone.
- **Hybrid** (0.75 emb. + 0.25 NTL) beats either.
- Largest gains in countries with **weak statistical capacity**.
- *NTL still wins on short-run GDP growth.*



Complementary, not competing.

Embeddings = structural levels.

NTL = short-run activity.

Three Honest Caveats Before Moving On

1. Construct validity

The embedding does not “know” your social-science construct. It learns visual patterns that co-vary with labels in the training data.

So transportability is never automatic: a fit in one setting can fail after institutional change, in a different country, or under landscape shift.

2. Interpretability

The 64 dimensions are not “urbanization,” “greenness,” . . . and 61 other nameable constructs.

Each is a *latent summary* — predictive in combination, but not individually nameable. Predictive leverage, not always mechanistic explanation.

3. Spatial bias

Training data are not uniformly distributed: the Global North is over-represented; cloud cover, sensor density, and biome coverage all vary.

Performance is uneven — and the unevenness correlates with wealth, state capacity, and geopolitical attention.

None of these are fatal. They define the **honest scope** of the tool today, and they set the research agenda for the next few years.

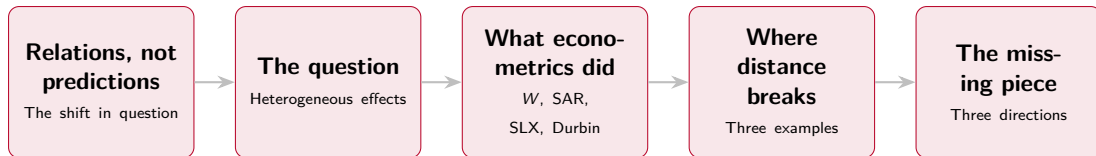
Where this leaves us, in three lines.

- 1 The applied literature on earth embeddings in social science is **thin**. We have a handful of computer-science benchmarks, no political-science papers, and a clear menu of substantive topics waiting to be tried.
- 2 The natural place to start is **physically grounded outcomes** — public goods, infrastructure, urban form — because that is what the embedding is built to see, and Gong et al. already show this is exactly where it works best.
- 3 Beyond that, whether earth embeddings help is **an empirical question**, one outcome at a time. *Building those benchmarks is the near-term agenda for the field.*

But prediction is not the whole of social science. Political scientists ask about **effects of X on Y** — and that shifts the question entirely. *Part 3.*

Part 3: A Methodological Opening

From prediction to relation: a new ingredient for spatial estimation



Question answered in Part 3:

Could earth embeddings give spatial estimation an ingredient it has been missing for thirty years?

Social Science Asks About Relations, Not Predictions

Part 2 world: prediction

Treat embeddings as rich covariates. See how far they carry a supervised model toward a held-out target.

$$\hat{y} = f(\mathbf{x}, \mathbf{e})$$

Useful — but not what political scientists *usually* mean when they ask about geography.

Part 3 world: relation

Hold everything else fixed and ask what X does to Y for unit i .

$$\frac{\partial Y_i}{\partial X_i} = \beta_i$$

Aid $\rightarrow$ schooling. Minimum wage $\rightarrow$ employment.
Civil war shock $\rightarrow$ market disruption.

The question driving Part 3:

Can the effect of X on Y for unit i be heterogeneous in the geographic character of i itself?

The Conceptual Opening

The technical question: how do you operationalize “structurally different” in an equation?

Before:

- Distance
- Dummy variable
- Fixed effect
- Hand-crafted proxy (NTL)

None carries the *character of the place itself*. They carry identity, or position.

Now:

- A 64-dim vector per location
- Compact summary of built form, road network, land use, vegetation
- **Two places can be similar** in a way geography alone cannot capture

Before this technology existed, there was no good way to put “the visual, morphological character of a place” into an equation. *Now there is.*

What Econometrics Did with This Question

Spatial econometrics encodes geographic structure in a **single matrix**, $\mathbf{W}$, that says *who counts as a neighbor of whom*.

The matrix $\mathbf{W}$.

$$\mathbf{W} = [w_{ij}]_{N \times N}, \quad w_{ii} = 0$$

- $w_{ij} \geq 0$: how strongly j counts as a neighbor of i .
- Diagonal is zero: a unit is not its own neighbor.
- Rows row-standardized, $\sum_j w_{ij} = 1$, so $\mathbf{W}$ delivers *averages*.

What you do with it: the spatial lag.

Multiplying any vector $\mathbf{z}$ by $\mathbf{W}$ replaces each entry with a **weighted average of its neighbors**:

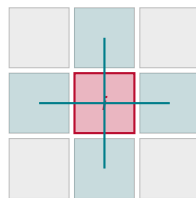
$$(\mathbf{W}\mathbf{z})_i = \sum_j w_{ij} z_j = \bar{z}_{\text{neighbors of } i}$$

Plug the outcome $\mathbf{y}$ in $\rightarrow$ “my neighbors’ y ”. Plug a covariate $\mathbf{x}$ in $\rightarrow$ “my neighbors’ x ”. Every spatial model is built from this one operation.

Why bother? **When unobserved factors cluster in space**, residuals become correlated, IID assumptions fail, and standard errors get misleading. $\mathbf{W}$ is how the model writes that structure down.

How W Is Usually Specified — The Short Menu

- **Distance (Euclidean).** $w_{ij} = f(d_{ij})$ (farther pairs get lower weight).
- **Inverse Euclidean distance.** $w_{ij} = 1/d_{ij}$ (or $1/d_{ij}^2$, $e^{-\alpha d_{ij}}$).
- **Contiguity.** $w_{ij}=1$ if i, j share a border.
- **k -nearest neighbors.** Each unit connected to its k closest (by Euclidean d_{ij}).
- **Distance bands.** $w_{ij}=1$ if $d_{ij} < r$.



contiguity W

Distance is still the primitive: either directly through d_{ij} , or coarsened into contiguity.

SAR, SLX, Durbin — The Three Families

SAR

Spatial Autoregressive. Outcome depends on a lag of itself.

$$\mathbf{y} = \rho \mathbf{W}\mathbf{y} + \mathbf{X}\beta + \varepsilon$$

y_i depends on i 's covariates, plus a weighted average of *neighbors'* y .

SLX

Spatial Lag of $\mathbf{X}$. Outcome depends on a lag of covariates.

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{W}\mathbf{X}\gamma + \varepsilon$$

y_i depends on i 's covariates, plus a weighted average of *neighbors'* X .

Durbin

Combines both. The general case.

$$\mathbf{y} = \rho \mathbf{W}\mathbf{y} + \mathbf{X}\beta + \mathbf{W}\mathbf{X}\gamma + \varepsilon$$

Both lags in one model.

Estimation is mature — MLE, Bayesian, GMM, 2SLS. The real bottleneck is upstream: **many ways to estimate, but almost always with a distance-based $\mathbf{W}$ at best.**

More detail: [W from scratch](#) ▪ [SAR](#) ▪ [SLX](#) ▪ [Durbin](#)

Who Has Pushed Back

The field has argued that distance-only W is too blunt, and that W should be *estimated* rather than pre-declared.

Critique of blunt distance.

- **Neumayer & Plümpert (2016)**, “W”. Distance-or-contiguity W tells us little more than that places get more dissimilar farther away.
- **Beck, Gleditsch & Beardsley (2006)**. Trade-based W for political economy.
- **Cook, Hays & Franzese (2022)**. Of 201 cross-country panel articles, only 12 model both temporal and spatial dependence.
- **Wimpy, Whitten & Williams (2021)**. SLX is underused relative to SAR.

More detail: [estimating \$W\$ from data \(paper map\)](#)

Estimating W from data.

- **Tan, Kesina & Elhorst (2025)**. Parameterize decay: $w_{ij} \propto d_{ij}^{-\alpha}$; estimate α jointly with β .
- **Lam & Souza (2020)**. W as weighted combination of candidates + adaptive LASSO.
- **Zhang & Yu (2018)**. Model averaging over candidate W s.
- **Qu & Lee (2015)**. Endogenous W .

The literature has been asking, implicitly, for a **non-geographic, non-flow-based candidate basis**. It just did not have one.

Where Distance Stops Making Sense

Distance-based W is correct for a class of **data-generating processes** (DGPs) — the mechanisms by which data actually arise — where spillovers really do run through proximity (pollution, disease).

The **harder class of DGPs** is the one where spatial dependence does not run through proximity at all.

Industrial river

Two sides of
the Mississippi.

Downtown STL vs. East Saint Louis.

Visually adjacent; a century
of zoning and race apart.

Mountain vs. valley

Dense urban fabric below.

Valley of Mexico City
and the ridges above it.

Rural communities
on the peaks. Almost
nothing in common.

Wall between worlds

Gated community next
to informal settlement.

Rocinha vs. São Conrado, Rio.

A single city block
apart. Disjoint mar-
kets and institutions.

In each case, the units have **structural** characteristics a spatial model would want to use — but that a classical distance-based W is blind to.

Earth Embeddings — The Missing Piece

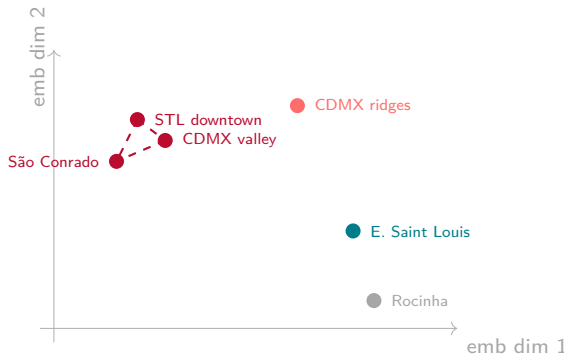
Earth embeddings compress the observable character of a place into a vector. For the first time, we have a **data-driven representation of what a place is like**.

Similarity, not distance.

$$s_{ij} = \cos(\mathbf{e}_i, \mathbf{e}_j) = \frac{\mathbf{e}_i \cdot \mathbf{e}_j}{\|\mathbf{e}_i\| \|\mathbf{e}_j\|}$$

A continuous measure of how much two places *resemble* each other in built form, road network, land use, and texture — regardless of how far apart they sit on a map.

An embedding-derived W — w_{ij} rising with similarity — is exactly the **non-geographic candidate basis** the literature has been asking for.



Three dense, formally planned urban cores on three continents (red) cluster together; the other places, structurally different, sit elsewhere. São Conrado clusters with STL downtown and the CDMX valley — not with its geographic neighbor Rocinha.

Three Directions for Future Work

Direction 1

Hybrid W

$$\mathbf{W} = f(d, s)$$

Functional form and parameters estimated jointly with the regression.

Ambitious version: retrain the location encoder so the vector is *native* to spatial estimation.

Direction 2

Add an S matrix

$$\mathbf{y} = \rho \mathbf{W}\mathbf{y} + \lambda \mathbf{S}\mathbf{y} + \mathbf{X}\beta$$

Both lags carry their own coefficient. The estimator *reports* how much is geography vs. similarity.

Most incremental, most testable — likely starting point.

Direction 3

Regularize W by similarity

Keep geographic $\mathbf{W}$; reweight entries by embedding similarity. Two units close in distance but far in embedding have w_{ij} *downweighted*. Fits inside the penalized- $\mathbf{W}$ framework of Lam & Souza (2020).

Two arms of open research.

Conceptual: which DGPs actually warrant a similarity-based $\mathbf{W}$? *Methodological:* how should similarity enter the estimator?

Earth embeddings give spatial econometrics
something it has not had before:
a non-distance, non-flow, non-hand-crafted
candidate basis for the similarity structure W .

The claim.

The question is *worth asking*, and the raw material for asking it *now exists*.

The frame.

The conceptual frontier has
a real candidate answer for
the first time in thirty years.

Conclusion

Three arguments, one technology

Three Arguments, One Technology

Part 1 — How

Earth embeddings compress the observable character of a place into a short vector, produced by Vision Transformers trained with self-supervised objectives on satellite imagery. The machinery is mature, pre-computed, and accessible by coordinate query.

Part 2 — What

The “what” is prediction tasks. Physically grounded outcomes are the natural first case because they are visible from above. Beyond that first corner, usefulness is empirical — outcome by outcome.

Part 3 — Next

On the estimation side, embeddings give spatial econometrics something it has not had before: a *non-distance* candidate basis for the similarity structure W . Two open arms of work: which DGPs warrant similarity-based W , and how similarity should enter the estimator.

Three Arguments, One Technology

Part 1 — How

Earth embeddings compress the observable character of a place into a short vector, produced by Vision Transformers trained with self-supervised objectives on satellite imagery. The machinery is mature, pre-computed, and accessible by coordinate query.

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The “what” is prediction tasks. Physically grounded outcomes are the natural first case because they are visible from above. Beyond that first corner, usefulness is empirical — outcome by outcome.

Part 3 — Next

On the estimation side, embeddings give spatial econometrics something it has not had before: a *non-distance* candidate basis for the similarity structure W . Two open arms of work: which DGPs warrant similarity-based W , and how similarity should enter the estimator.

Roads, buildings, vegetation, land use:
the full geographic character of a place now fits in a few dozen numbers.

What political science does with that is an open question.

One Technology, Three Literatures

Earth embeddings are a new data layer useful for Political Science.

The contribution of this talk is to read three literatures together —

Multimodal representation learning

Where the technology was built.

Economic and Political measurement

Where it can be better used.

Spatial econometrics and political methods

Where it could go next.

Part 1 (How) — next.

Improve geospatial foundation-model architectures (multi-sensor fusion, multimodal learning), while auditing hidden representation biases and domain shift.

Part 2 (What) — next.

Build systematic benchmarks for political-science outcomes: where embeddings are accurate, where they fail, and how they compare to classical proxies (NTL, census, administrative data).

Part 3 (Estimation) — next.

Characterize DGPs in which similarity between units is substantively meaningful. Develop methods that integrate that similarity into spatial estimation together with conventional geographic structure.

Thank you.

Questions?

Technical Deep-Dive

ViT math, attention, training losses, how each model actually produces a vector

▷ [Jump](#)

Anticipated Q&A

Technical and non-technical questions, with prepared responses

▷ [Jump](#)

Papers by Section

Core papers and supporting literature organized by Part 1, 2, 3

▷ [Jump](#)

- ▷ Vision Transformers: attention, patches, the CLS token
- ▷ Masked autoencoding: the self-supervised reconstruction loss
- ▷ Contrastive learning (InfoNCE): positives, negatives, temperature
- ▷ SatCLIP (Klemmer et al. 2025): coords $\leftrightarrow$ Sentinel-2, global coverage
- ▷ GeoCLIP (Vivanco Cepeda et al. 2023): coords $\leftrightarrow$ ground photos
- ▷ AlphaEarth Foundations (Brown et al. 2025): architecture + training trio
- ▷ Model ecosystem at a glance
- ▷ **W from scratch** — five slides, step by step
- ▷ SAR: spatial lag of the outcome
- ▷ SLX: spatial lag of covariates
- ▷ Durbin / SDM: both lags
- ▷ Estimating **W** from data (literature map)

Pipeline (Dosovitskiy et al., ICLR 2021)

- 1 Split image into fixed patches (typically 16×16 px)
- 2 Linear projection: patch $\mapsto$ vector (patch embedding)
- 3 Add learned **positional encoding**
- 4 Stack of L transformer blocks: each is **self-attention** + MLP + residuals
- 5 Output: contextualized patch tokens + [CLS] token
- 6 [CLS] (or pooled tokens) is “the embedding”

Self-attention, schematically

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^{\top}}{\sqrt{d_k}}\right) V$$

For each patch, compute a *query* vector Q_i , match it against every other patch's *key* K_j , and output a weighted sum of their *value* vectors V_j . The model learns which patches should attend to which.

Why it matters for geography: self-attention lets a patch that could be, say, a parking lot borrow evidence from distant patches in the same image (roads, buildings, vegetation) without a fixed spatial stencil.

Masked Autoencoding (MAE)

Training procedure

- 1 Randomly mask a large fraction ($\sim 75\%$) of patches
- 2 Encoder sees only the *visible* patches
- 3 Decoder tries to reconstruct the hidden patches
- 4 Loss: mean squared error on the masked pixels

$$\mathcal{L}_{\text{MAE}} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \|x_i - \hat{x}_i\|^2$$

where $\mathcal{M}$ is the set of masked patches and $\hat{x}_i$ is the decoder's reconstruction.

What the embedding ends up encoding To reconstruct hidden patches from visible ones, the encoder has to learn *what landscapes look like*: regularities of urban form, vegetation texture, road geometry, how buildings cluster. **Used by**

- **Clay Foundation model** (multi-sensor, resolution-agnostic)
- **Prithvi** (NASA / IBM, Hydrology / wildfires)

Originally proposed for images by He et al. (2022); the idea itself goes back to BERT's masked language modeling.

Contrastive Learning (CLIP-Style, InfoNCE)

What one minibatch looks like

- 1 Sample B **aligned** pairs (a_i, b_i) : same place, two modalities (e.g. lat/lon and the satellite chip centered there).
- 2 Two encoders f_a, f_b produce vectors $\mathbf{u}_i = f_a(a_i)$, $\mathbf{v}_i = f_b(b_i)$.
- 3 For anchor i , the **positive** is the matching index i . Every other $j \neq i$ in the batch is treated as a **negative** for that anchor (in-batch negatives).
- 4 Train so that $\text{sim}(\mathbf{u}_i, \mathbf{v}_i)$ is large and $\text{sim}(\mathbf{u}_i, \mathbf{v}_j)$ ($j \neq i$) is small, with cosine similarity or dot product on ℓ_2 -normalized vectors.

$$\mathcal{L}_i = -\log \frac{\exp(\text{sim}(\mathbf{u}_i, \mathbf{v}_i)/\tau)}{\sum_{j=1}^B \exp(\text{sim}(\mathbf{u}_i, \mathbf{v}_j)/\tau)}$$

τ (temperature) rescales logits: smaller $\tau \Rightarrow$ sharper

What you are *not* assuming The loss never mentions map distance. It only rewards cross-modal agreement at the same index. Still, nearby places often share land cover; distant places can share climate or urban form (Klemmer et al. 2025, Fig. 1). So embeddings can encode *ground conditions* that partly covary with geography without hard-coding d_{ij} .

Blueprint

- CLIP (Radford et al. 2021): image $\leftrightarrow$ text
- **SatCLIP**: coords $\leftrightarrow$ Sentinel-2 (▷)
- **GeoCLIP**: coords $\leftrightarrow$ ground photos (▷)

Klemmer et al. (AAAI 2025). Satellite Contrastive Location–Image Pretraining.

- **Goal:** a *global, general-purpose* location encoder: latitude/longitude $\mapsto$ vector, pretrained so that location embeddings match CNN/ViT summaries of Sentinel-2 chips at the same place (CLIP-style contrastive loss; Radford et al. 2021).
- **Why Sentinel-2 at pretraining:** uniform land coverage and multispectral signal for “ground conditions” worldwide—contrast with Flickr-based encoders that skew Western and street-level (see GeoCLIP slide).
- **Training data (paper):** S2-100K ($\sim 100k$ cloud-free Sentinel-2 images, Jan. 2021–May 2023, approximately uniform over landmass; Microsoft Planetary Computer).
- **Reported payoff:** improved performance on nine diverse downstream geospatial tasks vs. alternative location encoders; emphasis on geographic domain adaptation.

Access: open code/weights (Microsoft Research; see paper and repository). Embedding dimension depends on the released checkpoint (your deck uses one row in the table; keep it consistent with the checkpoint you actually load).

Vivanco Cepeda et al. (2023). Contrastive image and location encoders trained on **ground-level** Flickr-style photos (MediaEval Placing Task MP-16; see also discussion in Klemmer et al. 2025).

- **What it aligns:** natural photos $\leftrightarrow$ coordinates, not satellite pixels. Strong for geo-localization of *photographs*.
- **Geographic coverage caveat (literature):** photo density is uneven; Western countries are over-represented in MP-style collections. That shapes what the encoder “knows” about rare places.
- **Use for your story:** when the research design cares about built environment or human activity visible from the street, GeoCLIP is the closer modality. When you want land surface signal comparable across countries and years, SatCLIP or raster foundation models are usually closer.

Brown et al. (2025), Google DeepMind

- **Inputs:** multiple Earth-observation sources (optical, radar, lidar, climate layers, geo-tagged text) fused through source-specific encoders into a shared latent space; **STP** backbone mixes multi-resolution pathways (spatial-temporal processor; paper Fig. 2A–D).
- **Training trio (paper):** (i) teacher *video* embedding model with implicit decoders, (ii) student with shared architecture, (iii) text-alignment model—joint contrastive alignment across teacher/student and text (Fig. 2E).
- **Decoding losses:** outputs treated as mean direction of a **von Mises–Fisher** distribution; decoding samples that distribution and conditions on sensor geometry and timecodes

What you actually consume

The model weights are not the only artifact. For applied work, the practical object is often the **pre-computed embedding catalog** (e.g. Google Earth Engine):

- Global land coverage (paper: $\pm 82^\circ$ example mosaic), annual layers where provided
- ~ 10 m cells, query by point or polygon
- No need to retrain the foundation model for standard uses—you align your downstream task to the released field

Contrast with SatCLIP/GeoCLIP:

AlphaEarth is an *explicit* multisensor image $\rightarrow$ vector field with super-

Model Ecosystem, in Detail

Model	Family	Input	Training	Dim.	Access
AlphaEarth	Explicit	Sentinel-1/2 auxiliary layers	Supervised multi-task	64	Pre-computed, Google Earth Engine
Clay	Explicit	Multi-sensor imagery (drone to MODIS)	Masked autoencoding	varies (Matryoshka)	Open weights, self-hosted inference
SatCLIP	Implicit location encoder	Coords ↔ Sentinel-2 pairs	Contrastive (InfoNCE)	256	Open weights
GeoCLIP	Implicit location encoder	Coords ↔ ground-level photos	Contrastive (InfoNCE)	varies by backbone	Open weights
MOSAICS	(precursor)	Daytime imagery	Random features + ridge	fixed	Open code
Prithvi	Explicit	Harmonized Landsat Sentinel	MAE	varies	Open weights (NASA/IBM)

Rule of

thumb: *Explicit* extractors process imagery into vectors (you give them pixels). *Implicit* location encoders return vectors directly from coordinates (you give them a (lat, lon)). AlphaEarth is the most accessible today

W From Scratch — Roadmap of the Next Seven Slides

A step-by-step explanation of the spatial weights matrix $\mathbf{W}$, aimed at building intuition before the formal SAR/SLX/Durbin slides.

- ▷ **Step 1.** The picture: 4 cities, who is whose neighbor?
- ▷ **Step 2.** Writing it as a matrix
- ▷ **Step 3.** Row-standardize to get *averages*
- ▷ **Step 4.** The spatial lag: $\mathbf{W}\mathbf{y}$ as “my neighbors’ average y ”
- ▷ **Step 5.** What changes when $\mathbf{W}\mathbf{y}$ enters the equation: β , $\mathbf{u}$, bias
- ▷ **Step 5b.** What this does to estimates, SEs, and interpretation
- ▷ **Step 6.** How people usually choose $\mathbf{W}$, and why it matters

The whole point: every spatial econometric model uses the same machine, $\mathbf{W}\mathbf{z}$. Once you understand that one operation, you understand SAR, SLX, and Durbin.

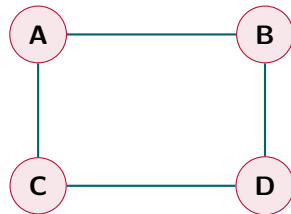
Step 1 — The Picture: Who Is Whose Neighbor?

Toy example. Four cities, A , B , C , D .

- A shares a border with B and C .
- B shares a border with A and D .
- C shares a border with A and D .
- D shares a border with B and C .

Question: *when we estimate effects on A , who counts as A 's neighbor and how strongly?*

W is the answer. It is just a way of writing “who counts” down as a table.



Each line says “these two are neighbors.” No line, no neighborhood relationship.

Step 2 — Writing It as a Matrix

Take the picture and turn it into a 4×4 table. Put a 1 where two cities are neighbors, a 0 otherwise. The diagonal is always 0 (a city is not its own neighbor).

$$\mathbf{W}^{\text{raw}} = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

Rows and columns indexed by A, B, C, D . Read row A across: A has B and C as neighbors.

Three rules of every $\mathbf{W}$.

- 1 $w_{ii} = 0$ on the diagonal.
- 2 $w_{ij} \geq 0$ everywhere else.
- 3 Whatever “neighbor” *means* (border, distance, similarity, ...) is hidden inside how you fill the off-diagonal entries.

Punchline. $\mathbf{W}$ is just *your theory of neighborhood* written as a table.

Step 3 — Row-Standardize to Get *Averages*

Divide each row by its sum so that every row of **W** adds up to 1. Now the entries are *shares*, not yes/no flags.

$$\mathbf{W} = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}$$

Every row sums to 1. *This is the **W** we actually use.*

Why we standardize.

- Without it, units with many neighbors get weighted heavier just because they have more friends.
- With it, **Wz** becomes a **weighted average**, not a sum — exactly the quantity we want when we say “what my neighborhood looks like”.

After this step, **W** is a machine that takes a variable **z** and gives back, for each unit, **the average of z over its neighbors**.

Step 4 — The Spatial Lag Wy

The only operation any spatial model uses. You take some variable on the four cities and multiply by **W**. Out comes *the same variable, but each entry is now the average over the unit's neighbors*.

Concrete: suppose income is

$$\mathbf{y} = \begin{pmatrix} 10 \\ 20 \\ 30 \\ 40 \end{pmatrix} \begin{matrix} A \\ B \\ C \\ D \end{matrix}$$

Then **Wy** is:

$$\mathbf{Wy} = \begin{pmatrix} \frac{1}{2}(20) + \frac{1}{2}(30) \\ \frac{1}{2}(10) + \frac{1}{2}(40) \\ \frac{1}{2}(10) + \frac{1}{2}(40) \\ \frac{1}{2}(20) + \frac{1}{2}(30) \end{pmatrix} = \begin{pmatrix} 25 \\ 25 \\ 25 \\ 25 \end{pmatrix}.$$

Read it column by column.

“A’s neighborhood income” = average of B’s and C’s incomes = 25.

“B’s neighborhood income” = average of A’s and D’s incomes = 25.

Etc.

Plug **y** in → “my neighbors’ **y**”.

Plug **x** in → “my neighbors’ **x**”.

That single step is what turns regression into spatial regression.

Step 5 — What Changes When Wy Enters the Equation

Same data, two specifications. What does adding the spatial lag actually *do* to β , to the error, and to the bias?

Without W (plain OLS)

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{u}$$

- If outcomes truly depend on neighbors' outcomes, $\mathbf{u}$ secretly contains $\rho\mathbf{W}\mathbf{y}$.
- Spatially clustered $\mathbf{X} \Rightarrow$ that hidden $\mathbf{W}\mathbf{y}$ correlates with $\mathbf{X} \Rightarrow \hat{\beta}_{\text{OLS}}$ is **biased** (omitted-variable bias, spatially flavored).
- $\mathbf{u}$ is **spatially autocorrelated**: standard errors too small, t -stats too big, false significance.

With W (e.g. SAR)

$$\mathbf{y} = \rho\mathbf{W}\mathbf{y} + \mathbf{X}\beta + \epsilon$$

- β now means **own effect of X holding neighbors' y fixed**.
- ρ explicitly carries the **spillover**; it is no longer hiding inside $\mathbf{u}$.
- ϵ is the **truly idiosyncratic** part: standard errors stop lying.

Bottom line: adding $\mathbf{W}\mathbf{y}$ moves spillovers out of the residual and into an explicit parameter, ρ .

Step 5b — What This Does to Estimates, Bias, and Errors

- **What happens to β ?** In SAR-type models, β is an *own-unit* effect conditional on neighbors' outcomes.
- **Where do spillovers go?** They get their own parameter, ρ , instead of being hidden in the error term.
- **What happens to the error?** It is closer to truly idiosyncratic noise, so standard errors are less likely to be artificially small.
- **How are effects reported?** Effects split into *direct* and *indirect* components (feedback through the spatial multiplier).

Actual effect of adding W : less omitted-spillover bias in β , cleaner errors, and an explicit decomposition of effects instead of one pooled slope.

Caveat: if W is misspecified, $\hat{\rho}$ absorbs that mistake and $\hat{\beta}$ can still be biased.

Step 6 — How W Is Usually Built (and Why It Matters)

In real applications, the off-diagonal entries are filled using one of a short menu. All of them are some function of geographic distance.

The standard menu.

- **Contiguity.** $w_{ij}=1$ if i and j share a border (the toy example above).
- **k -nearest neighbors.** Each unit is connected to its k closest others.
- **Inverse distance.** $w_{ij} = 1/d_{ij}$ (or $1/d_{ij}^2$, $e^{-\alpha d_{ij}}$).
- **Distance bands.** $w_{ij}=1$ if $d_{ij} < r$, 0 otherwise.

Different wrappers, but in every case the input is the same: **geographic distance**.

Why this matters.

Two specifications of **W** on the *same* data can give different point estimates, different standard errors, different signs.

The choice of W is a substantive modeling decision — not a technical default. And in applied work, it is almost always made *by distance alone*.

*This is exactly the slot Part 3 of the talk wants to fill: an embedding-based, similarity-driven candidate basis for **W** — a non-distance way of writing down what “neighbor” means.*

Spatial autoregressive (SAR) / spatial lag model. The outcome at i depends on a weighted average of *neighbors' outcomes*, creating simultaneity unless $\rho = 0$.

$$\mathbf{y} = \rho \mathbf{W}\mathbf{y} + \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

- **Interpretation:** diffusion, spillovers, or network dependence in levels expressed through $\mathbf{W}\mathbf{y}$.
- **Estimation note:** likelihood / GMM / IV-style strategies address the $\mathbf{W}\mathbf{y}$ endogeneity; software stacks are standard (e.g. `spdep`, spatial econometrics texts: Anselin 1988; Elhorst 2014; Lesage & Pace).
- **Link to your argument:** once $\mathbf{W}$ is wrong, ρ absorbs misspecified distance—your Part 3 motivation.

Spatial lag of X (SLX). Neighbors' covariates enter as extra regressors; no $\mathbf{W}\mathbf{y}$ term, so OLS is often the starting estimator when the spatial weights attach only to exogenous $\mathbf{X}$.

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{WX}\gamma + \varepsilon$$

- **Interpretation:** contextual effects, spatial externalities in observables, “what my neighbors look like” channels.
- **Political methodology angle:** Wimpy, Whitten & Williams (2021) argue SLX is under-used relative to SAR even when the substantive story is about $\mathbf{WX}$ rather than autoregressive levels.

Spatial Durbin / general nesting. Combines SAR and SLX:

$$\mathbf{y} = \rho \mathbf{W}\mathbf{y} + \mathbf{X}\boldsymbol{\beta} + \mathbf{W}\mathbf{X}\boldsymbol{\gamma} + \boldsymbol{\varepsilon}$$

- **Nesting:** set $\rho = 0$ for SLX-only; set $\boldsymbol{\gamma} = \mathbf{0}$ for SAR on exogenous $\mathbf{X}$ only.
- **Practical point:** richer propagation stories, more parameters, and the same $\mathbf{W}$ sensitivity as the simpler members of the family.

Complements the “Who has pushed back” slide in Part 3: these papers keep classical spatial structure but let the data choose decay or weights instead of fixing contiguity only.

- **Tan, Kesina & Elhorst (2025)**. Parametric distance decay inside w_{ij} ; estimate decay jointly with regression parameters instead of fixing α ex ante.
- **Lam & Souza (2020)**. Write $\mathbf{W}$ as a weighted combination of candidate matrices and use adaptive LASSO-style shrinkage to learn sparse effective neighbors.
- **Zhang & Yu (2018)**. Model averaging over a set of candidate $\mathbf{W}$ s when theory does not pin down a unique spatial structure.
- **Qu & Lee (2015)**. Treat spatial weights as endogenous objects—relevant when $\mathbf{W}$ itself is built from outcomes or strategic interactions.

Connective tissue: all are still *matrix*-spatial approaches; embedding-based similarity is a different candidate basis for w_{ij} , not a substitute for thinking about identification.

Stage 1: NTL Mapper (transfer learning)

- Inputs: 64-dim AlphaEarth embeddings
- Target: VIIRS nighttime light intensity
- Net: FC(64→128→64→32→1)
- *Truncate* at the 32-dim hidden layer
- Output: “income-aware” 32-dim representation
 $\mathbf{z}_i = g_\theta(\mathbf{x}_i)$

Stage 2: GDP Mapper (hierarchical)

- Per-cell: $\mathbf{v}_i = p_\psi(\mathbf{z}_i) \in \mathbb{R}^{16}$
- Country-level sum-pool: $\mathbf{s}_c = \sum_{i \in c} \mathbf{v}_i$
- Stage 2 predictor: $\hat{y}_c^{(\text{EMB})} = h_\phi(\mathbf{s}_c)$

Hybrid

$$\hat{y}_c^{(\text{Hybrid})} = \lambda \hat{y}_c^{(\text{EMB})} + (1 - \lambda) \hat{y}_c^{(\text{NTL})}$$

Key numbers

- GDP levels (cross-sectional, MSE):
EMB 0.504, NTL 0.602, Hybrid ($\lambda=0.75$) 0.459
- GDP growth (within-country, MSE):
EMB 0.0084, NTL 0.0075 (*NTL wins short-run*)

Error-decomposition framework (Eqs. 9–15)

- $\varepsilon^{(\text{stat})}$: statistical-system noise
- $\varepsilon^{(\text{sat})}$: satellite observability noise
- Predicts: EMB wins where SPI is low (weak stats)

Technical questions

- ▷ Why Vision Transformers, not CNNs?
- ▷ Why 64 dimensions?
- ▷ Which training objective is “right”?
- ▷ How is this different from MOSAIKS?
- ▷ Interpretability?
- ▷ Spatial bias in training data?

Non-technical / substantive

- ▷ Why political science specifically?
- ▷ Isn't this just fancy remote sensing?
- ▷ Why not just use nighttime lights?
- ▷ Where is the causal identification?
- ▷ What is the originality claim in Part 3?
- ▷ What would a failed version of this agenda look like?

Q: Why Vision Transformers, not CNNs?

Short answer. Self-attention is better at integrating long-range spatial context in a single imagery patch than hierarchical convolutions. CNNs remain viable (ResNet/EfficientNet are still used for some geospatial tasks), but the transformer has become the default in recent foundation models because:

- **Global receptive field from layer 1.** A patch representing a river can attend directly to the patches representing its banks, its estuary, its floodplain — without waiting for deeper convolutional layers to widen the receptive field.
- **Scales cleanly with data.** Transformers keep improving with larger datasets in a way CNNs plateau (Dosovitskiy et al. 2021).
- **Unified framework across modalities.** The same attention machinery that handles satellite patches handles radar, text, auxiliary tabular data — which matters for **fusion** (second Klemmer function).

Honest caveat. At small spatial scales, with small datasets, a well-tuned CNN is often competitive. The advantage is asymptotic in data and in spatial context.

Q: Why 64 dimensions?

Short answer. 64 is an *engineering* choice by Google DeepMind for AlphaEarth, not a theoretical floor. Other models use 256 (SatCLIP), 512, or variable (Clay's Matryoshka). The relevant question is not “why that number” but *what it buys and what it costs*.

What small d buys

- Lower storage and bandwidth (global catalog at 10m res is already petabyte-scale)
- Faster downstream models
- Less risk of overfitting in small-sample applied work
- Forces the backbone to prioritize features jointly useful for all supervised tasks

What small d costs

- Less headroom for fine-grained distinctions
- If the outcome depends on features orthogonal to the training tasks, small d squeezes them out

Empirical data point: Gong et al. (2026) find AlphaEarth's 64-dim vectors beat Prithvi and Clay's higher-dim vectors on 14 urban indicators — suggesting 64 is enough for most built-environment signals.

Q: Which training objective is “right”?

Short answer. None is universally right. The objective should match the downstream research question.

- **MAE** learns visual / textural regularities. Strong when the outcome depends on landscape structure per se (land cover, vegetation, built-up intensity). Less targeted toward any specific downstream semantic.
- **Contrastive** learns location identity from recurring observable patterns. Strong when the research question is fundamentally about *similarity* between places (SatCLIP treats two industrial zones on different continents as close). This is the objective that matters most for the Part 3 proposal.
- **Supervised multi-task** embeds the targets you tell it to embed. Strong when you know in advance the bundle of things you want to recover, and you are willing to accept the bias of those targets shaping the vector.

Practical recommendation for political scientists today: start with AlphaEarth for measurement work (accessible, pre-computed, strong on built environment); start with SatCLIP if the research question turns on structural similarity between places (e.g., the Part 3 direction).

Q: How is this different from MOSAIKS?

Short answer. MOSAIKS is the direct precursor; it proved that a single feature set can predict many outcomes. But it uses *random* convolutional features combined with ridge regression, rather than a *learned* representation trained with a principled objective. Klemmer et al. (2025) explicitly exclude it from the “earth embeddings” category on this basis.

	MOSAIKS (Rolf et al. 2021)	Earth embeddings
Feature origin	Random convolutional filters	Learned via self-supervision / multi-task
Training objective	<i>None</i> (features are frozen random projections)	MAE, contrastive, or supervised multi-task
Geometry of space	No meaningful geometry (random)	Geometry reflects similarity, geography, or both
Usable as “similarity” input?	No — random dims are not interpretable	Yes — cosine similarity is semantically meaningful
Contribution	Established that “one feature set, many outcomes” is	Extends this into a principled representation

Q: Aren't these black boxes? What about interpretability?

Short answer. Yes, they are black boxes in the sense that individual dimensions have no a priori semantic content. That is a real cost, and the paper is upfront about it in Section 5.2.

What's being done in the literature (that political scientists will *inherit*, not solve):

- **Probing classifiers.** Train a small supervised model to predict an interpretable quantity (land cover, population density, imperviousness) from embedding dimensions. What can be probed back out is a proxy for what the embedding “knows.”
- **Attention visualization.** For transformer-based models, visualize which patches of imagery contribute most to a given embedding, giving a spatial localization of what the embedding represents at that location.
- **Embedding Cards (Klemmer et al. 2025).** A documentation standard for what a given model's vectors encode, analogous to Model Cards in NLP.

This is a limitation that applied political-science work will live with for the foreseeable future. The question is not whether embeddings are fully interpretable — they are not — but whether the gain in coverage and signal is worth it. In settings where conventional data are missing, it clearly is.

Q: What about spatial bias in training data?

Short answer. Real concern, documented, and partially mitigable. Purohit et al. (2025) quantify the pattern: embeddings are more informative in well-represented regions (Europe, North America, East Asia, urban areas) than in under-represented ones (Sub-Saharan Africa, Central Asia, rural regions).

Why it matters especially for political science. Comparative work across diverse institutional contexts is precisely the setting where embeddings would be most valuable as a measurement technology, and also precisely the setting where their quality degrades. **What can be done**

- Report embedding quality by geography (akin to reporting standard errors by strata)
- Use residual diagnostics — does the embedding explain less variation in under-represented regions? If so, interpret coefficients accordingly.
- Prefer models explicitly trained with geographic balancing objectives (AlphaEarth's training corpus is more geographically balanced than most; Clay is explicitly tunable on new corpora).
- For cross-country work, combine embeddings with complementary proxies (NTL, census, auxiliary layers) to reduce reliance on any single signal.

Q: Why political science specifically?

Short answer. Two reasons, corresponding to Parts 2 and 3.

- **Measurement (Part 2):** Political science spends a lot of its empirical energy on public goods provision, state capacity, local infrastructure, conflict, and development outcomes — all of which leave physical traces that embeddings are built to see. Several of these questions are chronically constrained by data availability in low-income and conflict-affected settings, which is exactly where a queryable, coordinate-based representation is most valuable.
- **Methodology (Part 3):** Political methodology has been a major consumer of spatial econometric tools for thirty years. The discipline already has a relationship with the matrix W and its discontents. Earth embeddings offer a new ingredient for that discussion, and political methodologists are among the best positioned to think carefully about *when* similarity-based W makes substantive sense.

What I am *not* claiming. That this is uniquely political science's job. Economic geographers, sociologists, and development economists are already using many of these tools. The argument is that *our* discipline hasn't noticed, and that the methodological questions in Part 3 sit particularly comfortably in political methodology.

Q: Isn't this just fancy remote sensing?

Short answer. Not quite. The distinction matters. **What earlier-generation remote sensing is.** A pipeline: acquire imagery, select spectral bands, preprocess (cloud masking, radiometric correction), engineer task-specific features, train a model. Each step takes specialized expertise. Each new outcome requires rebuilding the pipeline. Burke et al. (2021, *Science*) document this as the bottleneck that kept satellite-based measurement from going mainstream. **What earth embeddings change.**

Pre-computation moves the expertise out of the applied researcher's workflow. The same 64-dim vector per location is useful across many outcomes and tasks without task-specific training. The applied step *is merge with a tabular dataset*, not *build a remote-sensing pipeline*.

The substantive difference is **generality** and **accessibility**, not magic. A political scientist who can write a `pandas.merge()` can now use a representation that previously required a remote-sensing team to produce.

Q: Why not just use nighttime lights?

Short answer. Use both. The evidence says they are *complementary*, not competing. **What NTL is good at**

- Short-run economic activity (Yue et al. 2026 find NTL beats embeddings on GDP *growth*)
- Very long time series (back to 1992 with DMSP-OLS; daily with VIIRS)
- Single, interpretable scalar

What embeddings are good at

- Cross-sectional levels and structural features (Yue et al. 2026: 16% lower MSE on GDP levels; gains largest in low-SPI countries)
- Much richer feature space — can represent land use, urban form, road networks, vegetation
- Generalizes across outcomes without task-specific pipelines

See also Gibson, Alimi, and Boe-Gibson (2025)'s critical review of NTL use in economics, which catalogs well-known limitations of luminosity as a single scalar (saturation, blooming, gas-flare confounding).

Q: Where is the causal identification?

Short answer. This paper is not a causal-estimation paper; it is a data-layer + methodology paper. Causal identification lives in whatever research design the embedding is plugged into. **For Part 2 (measurement).** Embeddings serve as covariates, controls, or as the dependent variable itself when measuring a physical outcome. Identification of any causal claim is inherited from the research design the applied paper uses, not from the embedding. **For Part 3 (spatial estimation).** Here the identification question is sharper and worth naming precisely:

- If W (or S) is pre-specified by the researcher on the basis of *pre-treatment* embeddings, SAR / SLX / Durbin identification goes through as usual.
- If S is built from post-treatment embeddings, then S is endogenous and the situation resembles Qu and Lee (2015): we need either an IV strategy, a structural-parameter approach, or a dynamic setup where the timing of S relative to Y pins identification down.
- Section 5.4 of the paper explicitly names this: embeddings are a data layer, not a research design. A well-specified embedding does not rescue a poorly identified model.

Q: Why is Part 3 an original contribution?

Short answer. The contribution is more modest than “nobody has ever thought of this.” It is: the raw material to *try* this only became usable within the last two years, and no peer-reviewed paper has yet proposed using earth embeddings to specify W . **What is *not* original.**

- The idea that W should reflect something other than distance (Neumayer & Plümper 2016; Beck, Gleditsch, Beardsley 2006).
- The idea that W should be estimated rather than pre-declared (Lam & Souza 2020; Tan et al. 2025; Qu & Lee 2015).
- The general enthusiasm for satellite-derived proxies in social science (Burke et al. 2021; Henderson et al. 2012).

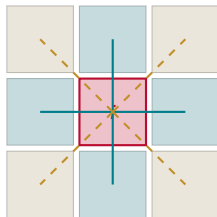
What is. Identifying that:

- 1 the estimating- W literature has been asking for a non-geographic, non-flow-based candidate basis it didn't have,
- 2 earth embeddings are exactly that candidate basis,
- 3 nobody has yet made the connection in print,
- 4 and sketching three concrete directions for the work.

Q: What is rook vs bishop vs queen contiguity in W ?

It is just a rule for who counts as a neighbor in polygon maps.

- **Rook contiguity:** neighbors share an **edge**.
- **Bishop contiguity:** neighbors share only a **corner** (diagonal touch).
- **Queen contiguity:** neighbors share either an edge **or** a corner (rook + bishop).



Solid lines: rook neighbors Dashed lines: bishop-only neighbors

In applied work, “bishop” alone is rare; the usual contrast is *rook* vs *queen*.

This choice changes the sparsity and meaning of W . **Rook** is stricter (more local), **queen** is denser. If your spillover mechanism plausibly crosses corners, queen is often preferred; if you want stronger locality, rook is cleaner.

Q: What would a failed version of this agenda look like?

An honest list of things that could go wrong.

- **Embedding similarity turns out not to add anything beyond geographic W after appropriate controls.** If similarity is essentially collinear with distance at the scales political scientists work at, the S matrix captures the same thing W already did.
- **Identification problems prove too sharp.** If the DGPs where similarity-based W is conceptually correct are also those where similarity is endogenous to the outcome, the program requires structural assumptions few substantive researchers are willing to make.
- **Construct drift.** Embedding pipelines change year to year. A research program whose core ingredient is a proprietary Google model is exposed to its revision cycle. Reproducibility concerns are real.
- **Geographic bias outweighs gains.** In exactly the under-represented settings where embeddings would be most valuable as a measurement tool, they may also be least reliable (Purohit et al. 2025).
- **The benchmark evidence doesn't generalize beyond the US.** Gong et al. (2026)'s 6-metro benchmark is a strong signal for the US urban setting. Extending the benchmark logic to comparative work is not trivial.

Crime incidence in Mexico City, 2020–2024.

- **Unit:** AGEB — the neighborhood-level polygon used in Mexican official statistics.
- **Outcome:** reported crime incidence per AGEB, per year (administrative records, used as the training target).
- **Inputs:** 2020 census variables at the block level, stacked with one AlphaEarth embedding per AGEB centroid.
- **Question:** does the embedding carry built-environment information that the sociodemographic covariates miss?

Why this is the right pilot.

- Neighborhood-scale outcome with a real *physical* signature (built form, street layout, density).
- The unit, AGEB, is fine enough that the embedding's spatial resolution is actually informative.
- Sits squarely inside the *earth-embedding* paradigm — not a generation-2 substitute.

This is the applied side of the dissertation-to-be, and it sets up the methodological argument in Part 3 (worked example slide).

- ▷ **Part 1 literature:** the embedding pipeline, ViTs, foundation models for Earth observation
- ▷ **Part 2 literature:** satellite-based measurement, benchmarks, applied work
- ▷ **Part 3 literature:** spatial econometrics, estimating W , political methodology

Core papers (★) are the ones discussed in depth during the presentation. Supporting papers are available for questions.

Core (★)

- ★ **Klemmer et al. (2025)**. “Earth Embeddings: Towards AI-centric representations of our planet.” *arXiv*. — The framework paper; formalizes the four functions.
- ★ **Brown et al. (2025)**. “AlphaEarth Foundations.” *arXiv 2507.22291*. — The supervised multi-task 64-dim model + Satellite Embedding Dataset.
- ★ **Dosovitskiy et al. (2021)**. “An Image is Worth 16×16 Words.” *ICLR*. — The Vision Transformer.

Supporting: concepts & precursors

- Mikolov et al. (2013): word2vec, the NLP antecedent.
- He et al. (2022): Masked Autoencoders in vision.
- Radford et al. (2021): CLIP (contrastive learning across modalities).

Supporting: other earth-embedding models

- Klemmer et al. (2025b): SatCLIP — contrastive coords ↔ Sentinel-2.
- Vivanco Cepeda et al. (2023): GeoCLIP — ground-level images.
- Clay Foundation model — MAE-trained, multi-sensor, Matryoshka.
- Prithvi (NASA/IBM) — MAE, hydrology/fires.

Supporting: precursor pipeline

- Rolf et al. (2021): MOSAIKS — random features, one feature set many outcomes.

Part 2 Papers — Measurement and Benchmarks

Core (★)

- ★ **Gong, Srivastava et al. (2026)**. “Earth embeddings reveal diverse urban signals from space.” *arXiv*. — Systematic benchmark, 14 urban indicators, 6 US metros.
- ★ **Yue, Zhao & Hu (2026)**. “Estimating Economic Activity from Satellite Embeddings.” *Applied Sciences*. — GDP pipeline, AlphaEarth + NTL transfer learning.
- ★ **Burke et al. (2021)**. “Using satellite imagery to understand and promote sustainable development.” *Science*. — Synthesis of generation-2 methods; identifies the bottleneck embeddings solve.

Supporting: generation-1 and generation-2

- Henderson, Storeygard & Weil (2012) — NTL → GDP.
- Donaldson & Storeygard (2016) — satellite data in economics (*JEP*).
- Jean et al. (2016) — CNN + NTL → African poverty.
- Gibson, Alimi & Boe-Gibson (2025) — critical review of NTL.

Supporting: applied political-science adjacent

- Sticher, Wegner & Pfeifle (2023): remote conflict monitoring (*PNAS Nexus*).
- Naghizadeh (2024): fire data + political violence (*R&P*).
- Edler et al. (2025): RS for human-rights documentation.
- Huang et al. (2024): public-service accessibility.
- Bell et al. (2023): RS at the science–policy interface.

Core (★)

- ★ **Neumayer & Plümper (2016)**. “ W .” *PSRM*. — The blunt critique of distance-only W .
- ★ **Cook, Hays & Franzese (2022)**. “STADL Up!” *APSR*. — TSCS spatiotemporal; documents the 12/201 finding.
- ★ **Tan, Kesina & Elhorst (2025)**. “Parameterizing spatial weight matrices.” *Political Analysis*. — Parameterized distance decay jointly estimated with regression.

Supporting: political methodology

- Beck, Gleditsch & Beardsley (2006): trade-based W .
- Wimpy, Whitten & Williams (2021): “X marks the spot” — SLX underused.
- Di Salvatore & Ruggeri (2021): spatial analysis in PS, review.

Supporting: estimating W

- Lam & Souza (2020): W as weighted combination + LASSO.
- Zhang & Yu (2018): model averaging over candidate W .
- Qu & Lee (2015): endogenous W .
- Krisztin & Piribauer (2022): Bayesian W estimation.